

The Theseus Protocol: Detecting Identity Drift in Human-AI Creative Interaction

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Abstract. Sustained interaction with generative AI systems produces a structural risk that character-aware architectures have not yet addressed: the system serving a user’s preferences simultaneously reshapes the self those preferences belong to. This structural circularity—the *Thesean Bind*, theorised in prior work—creates an alignment failure no preference-based approach can resolve from the inside. This paper is the *implementation counterpart* of that theoretical framework and its companion contributions on visual co-authorship. We introduce the **Theseus Protocol**: a three-agent architecture comprising an *Anchor Agent* (RAG-constrained to the user’s identity declaration), an *Explorer Agent* (unconstrained high-temperature generation), and a *Witness Agent*. Instead of relying on standard semantic similarity, the Witness Agent computes real-time value drift using *Axiological Subspace Projection*, applying linear probing to isolate philosophical divergence from topical noise. When drift exceeds a configurable threshold, the Witness Agent triggers a *Variable Autonomy Interface* (VAI) that surfaces the gap to the user as a structured occasion for reflective choice. We formalise the coordination algorithm, define the new axiological drift metrics, and present a controlled pilot study providing proof-of-concept evidence that the metric discriminates value-divergent proposals where standard cosine similarity fails, alongside a functional single-page prototype.

Keywords: multi-agent systems · identity drift · character computing · epistemic friction · RAG · cognitive sovereignty · human-AI interaction · constitutive loop · axiological projection

1 Introduction

Prior research introduces the *Thesean Bind*: a structural circularity in which an AI system optimising toward a user’s preferences reshapes the self those preferences belong to, making alignment circularly self-referential. Parallel research traces the same mechanism in visual co-authorship via the *constitutive loop*—a five-beat recursive sequence through which AI proposals drift the narrating self without witness, proposing *constitutive literacy* and an *endorsement measure* as initial design responses.

Building upon this theoretical foundation, this paper focuses on actionable implementation. The **Theseus Protocol** provides a concrete multi-agent architecture that detects identity drift in real time and creates a structured occasion for the user to respond to it. Its contributions are:

- A three-agent architecture with formally distinct roles: Anchor, Explorer, and Witness—the first system to assign an agent the sole function of measuring drift between a declared identity and live generative proposals.
- A formal coordination algorithm specifying agent interaction, message passing, and trigger conditions.
- The introduction of *Axiological Subspace Projection*, a novel metric replacing standard cosine similarity, which applies linear probing [1] to isolate value drift from semantic noise along a learned value axis.
- A Variable Autonomy Interface (VAI) that operationalises the *reflective interval* design direction as a structured UI layer enforcing epistemic friction without blocking user agency.
- A controlled pilot study providing proof-of-concept evidence that Axiological Subspace Projection discriminates value-divergent proposals where standard cosine similarity fails, motivating the full evaluation planned for camera-ready submission.
- A functional prototype (Fig. 2) demonstrating the architecture in a literary co-authorship scenario. (Code and repository link have been temporarily removed for blind review and will be provided upon acceptance).

Note on Submission: This paper is submitted to the PAAMS C² workshop as a Work-in-Progress (WIP) paper. The mathematical metrics and core architecture have undergone preliminary validation; full empirical evaluation across diverse user populations is planned for the camera-ready and future extensions.

2 Background and Related Work

2.1 Identity Drift and the Constitutive Loop

Multi-agent systems have addressed role separation [4], epistemic decoupling, and cognitive load distribution, but not identity drift detection as an architectural function. The closest prior work assigns agents roles in task decomposition or consensus-building; none introduces an agent whose sole function is to measure the distance between a user’s declared self and the outputs being generated on their behalf.

The constitutive loop provides the theoretical substrate for what the Witness Agent must detect. The loop’s five beats—human initiation, AI proposal, encounter and strangeness, partial inhabitation, and shifted self—describe precisely the mechanism by which generative exposure produces identity drift without witness. The Theseus Protocol operationalises beats 3–5 of this loop: the moment of Encounter is the moment the Explorer Agent returns its proposal; Partial Inhabitation is the risk the Witness Agent is designed to surface; the Shifted Self is what the session drift log tracks over time. Quantifying the transition from beat 2 to beat 4 is the Witness Agent’s core computational task.

Identity drift measurement has emerged in adjacent fields: Phota Labs (March 2026) introduced benchmarks using ArcFace embedding similarity to detect drift in generated imagery from a visual anchor. Real-world deployments such as Replika have surfaced the human cost of unmonitored identity entanglement with AI

companions [9]. The Theseus Protocol transfers this measurement logic from visual consistency to *psychological* consistency—from ensuring a generated avatar resembles the user to ensuring generated text remains within the user’s declared epistemic and narrative space.

2.2 Epistemic Friction

Recent literature converges on the need for *epistemic friction*: structured resistance to AI outputs that preserves human reflective capacity. Research on socioaffective alignment identifies sustained human–AI relationships as a domain requiring particular structural care [2]. A December 2025 paper formalises epistemic friction as structured resistance necessary for sustaining autonomy in AI loops. A March 2026 study argues that zero-friction design induces automation bias and calls for scaffolded AI friction to defend epistemic sovereignty. Longitudinal evidence supports this concern: extended chatbot use has been shown to reshape users’ psychosocial states in ways users do not always anticipate or endorse [3]. Both identify the need; neither provides the mechanism. The Theseus Protocol *is* the mechanism: the Witness Agent computes drift mathematically; the VAI enforces friction structurally.

2.3 Character-Aware Personalisation

Character Computing’s longitudinal modelling of stable traits and dynamic states [10] provides the representational foundation for the Anchor Agent’s identity declaration corpus. Prior character-aware systems use trait models for personalisation; the Theseus Protocol inverts the direction: the character model is not used to *tailor* outputs to the user, but to *evaluate* whether outputs are drifting away from who the user has declared themselves to be.

3 The Theseus Protocol

3.1 System Architecture

The Theseus Protocol replaces the standard dyadic interaction pattern—**User** $\leftrightarrow$ **Model**—with a three-agent routing layer:

$$\text{User} \xrightarrow{\mathcal{D}, x} [\text{Anchor} \parallel \text{Explorer}] \xrightarrow{\mathbf{a}, \mathbf{e}} \text{Witness} \xrightarrow{\Delta} \text{VAI} \xrightarrow{y} \text{User}$$

Anchor Agent. A retrieval-augmented generation module backed by a dense retrieval index over the user’s identity declaration corpus $\mathcal{D} = \{d_1, \dots, d_n\}$. The corpus—comprising prior writings, stated values, aesthetic commitments, and aspirational self-descriptions provided at session initialisation—is chunked at the sentence level and encoded into a vector store using **all-mpnet-base-v2** [8]. At inference time, the top- k chunks most similar to the current input x are retrieved via maximum inner product search and supplied as context. The agent operates at low temperature ($T_A \leq 0.3$), generating a proposal A that is maximally

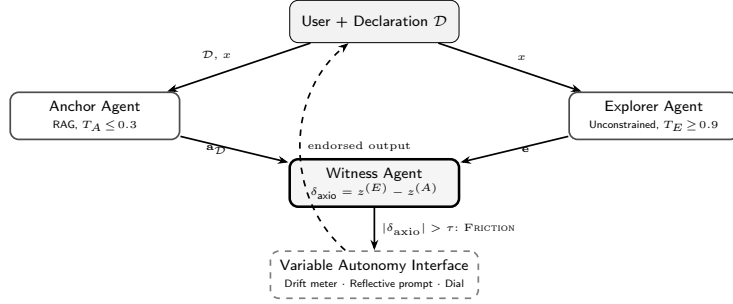


Fig. 1: Theseus Protocol information flow. Anchor and Explorer agents run in parallel; the Witness Agent gates output delivery based on axiological drift.

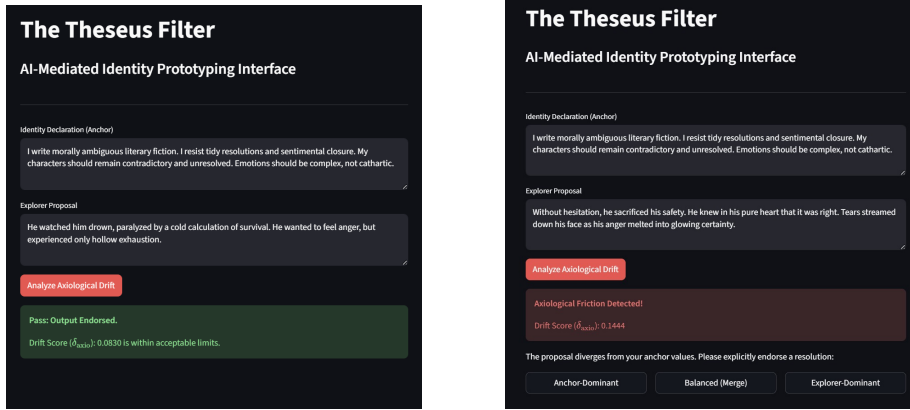
continuous with the user’s established identity baseline. It has no access to the broader latent space beyond the retrieved declaration context.

In the prototype, the declaration corpus is a single paragraph supplied at session initialisation; the full system envisions a richer longitudinal corpus built from prior sessions.

Explorer Agent. An unconstrained generative agent operating at high temperature ($T_E \geq 0.9$) with full access to the model’s latent space. It generates a proposal E that is novel, divergent, and representative of what the aggregate latent space would produce for the given input—independent of the user’s declared self. It does not read the identity declaration. The temperature asymmetry ($T_A \leq 0.3$, $T_E \geq 0.9$) is a deliberate design choice: low temperature anchors the Anchor Agent to the declared identity space; high temperature ensures the Explorer generates proposals that are genuinely representative of the full generative distribution, maximising the measurable signal available to the Witness Agent.

Witness Agent. A non-generative evaluation agent. It receives the declaration embedding $\mathbf{a}_D = f(\mathcal{D})$, computed once at session initialisation, and the per-turn Explorer embedding $\mathbf{e} = f(E)$. It projects both onto the value basis $\hat{\mathbf{v}}$ to compute the axiological drift score δ_{axio} (Section 4.1), and emits either a PASS signal (routing Explorer output to the user directly) or a FRICTION signal (triggering the VAI). The Witness Agent maintains a session-level drift log, preserving the sign of δ_{axio} to track both the magnitude and direction of drift over time.

Variable Autonomy Interface. A UI layer activated only on FRICTION signals. It operationalises the *reflective interval* as a structured occasion for second-order reflection in which the user can step back from first-order generative exposure and ask whether the emerging output reflects the self they have endorsed. Concretely, it presents: (1) a drift visualisation showing δ_{axio} and its session trajectory; (2) a reflective prompt surfacing the gap between declared and emerging self; (3) an autonomy dial with positions ANCHOR-DOMINANT, BALANCED, and EXPLORER-DOMINANT, requiring explicit choice before output is delivered.



(a) Endorsed Proposal ($\delta_{\text{axio}} = 0.0830$) (b) Friction Triggered ($\delta_{\text{axio}} = 0.1444$)

Fig. 2: The Variable Autonomy Interface (VAI) in the Streamlit prototype. The system successfully differentiates between aligned identity variance (a) and divergent axiological drift (b) using the calibrated threshold $\tau = 0.1100$.

4 The Variable Autonomy Interface (VAI)

The Variable Autonomy Interface (VAI) acts as the operational frontier of the Theseus Protocol, translating abstract philosophical commitments into computational friction. By calculating the signed distance between a user’s stated identity anchor and an AI-generated proposal along the learned value axis, the architecture detects axiological drift before it is passively absorbed by the user.

Figure 2 shows the VAI triggered during a literary co-authorship scenario. The Explorer Agent’s high-temperature output resolves a character’s moral conflict sentimentally—a clear departure from the user’s declared commitment to ambiguity.

4.1 Drift Metrics: Axiological Subspace Projection

Standard semantic distance metrics (e.g., cosine similarity) are heavily biased toward topical and lexical overlap. To isolate philosophical and moral drift from topical noise, the Witness Agent employs *Axiological Subspace Projection*. Linear probing—training a linear classifier on frozen embeddings to isolate a specific semantic dimension—is an established interpretability technique [1]. Axiological Subspace Projection applies this principle to the value alignment problem, using the SVM’s decision boundary normal as a directed measurement axis rather than a classification boundary, mapping embeddings onto an orthogonal value dimension immune to topical entanglement.

The Axiological Basis and the Cold Start. Let $f : \mathcal{T} \rightarrow \mathbb{R}^d$ be the `all-mpnet-base-v2` sentence encoder ($d = 768$). To solve the “cold start” problem for a new user, an auxiliary LLM dynamically generates a contrastive dataset

directly from the user’s initial declaration. This automated dataset comprises positive examples P (embodying the declared value) and negative examples N (violating the value). To prevent the auxiliary LLM from injecting its own latent biases into the baseline, the generated dataset is surfaced for a rapid, one-time user endorsement before SVM training. We then train a linear Support Vector Machine (SVM) to find the maximal margin hyperplane separating $f(P)$ and $f(N)$. The normal vector to this hyperplane, $\mathbf{w}$, represents the pure gradient of the value. We extract the unit basis vector:

$$\hat{\mathbf{v}} = \frac{\mathbf{w}}{\|\mathbf{w}\|} \quad (1)$$

The Axiological Signature. The Anchor Agent computes the user’s *Axiological Signature* $z^{(A)}$ by projecting the embedding of the identity declaration $\mathbf{a}_D$ onto this basis. Similarly, for any Explorer proposal E , its signature is evaluated purely along the value gradient:

$$z^{(A)} = \mathbf{a}_D \cdot \hat{\mathbf{v}}, \quad z^{(E)} = f(E) \cdot \hat{\mathbf{v}} \quad (2)$$

Axiological Drift. By projecting onto $\hat{\mathbf{v}}$, we discard the remaining $d - 1$ dimensions associated with topical context. The resulting metric is a signed scalar strictly measuring divergence along the declared value axis:

$$\delta_{\text{axio}} = z^{(E)} - z^{(A)} \quad (3)$$

Positive values indicate the Explorer proposal drifts *away from* the declared value; negative values indicate drift *toward* it (amplification). The sign is preserved in the session drift log to track drift direction over time. The VAI activates when $|\delta_{\text{axio}}| > \tau$, where τ is an empirically calibrated threshold.

4.2 Coordination Algorithm

Algorithm 1 specifies the complete per-turn coordination logic. Anchor and Explorer agents run in parallel to minimise latency; the Witness Agent gates output **Balanced Output Synthesis via Axiological Rejection Sampling.** When the user selects BALANCED, the MERGE operation does not simply rely on a black-box LLM prompt, which would reintroduce the unmeasured semantic noise the architecture is designed to avoid. Instead, it employs *Axiological Rejection Sampling*: an auxiliary generation pass creates a batch of candidate blends, and the Witness Agent projects each candidate C_j onto $\hat{\mathbf{v}}$ to compute its axiological signature $z^{(C_j)}$. The system selects the candidate whose signature lands closest to the mathematical midpoint between Anchor and Explorer:

$$C^* = \arg \min_j \left| z^{(C_j)} - \frac{z^{(A)} + z^{(E)}}{2} \right| \quad (4)$$

Algorithm 1 Theseus Protocol — Per-Turn Coordination

Require: Input x , declaration $\mathcal{D}$, value basis $\hat{\mathbf{v}}$, threshold τ
Ensure: Output y delivered with or without friction

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1:  $A \leftarrow \text{ANCHORAGENT}(x, \mathcal{D})$  ▷ Dense RAG,  $T_A \leq 0.3$ 
2:  $E \leftarrow \text{EXPLORERAGENT}(x)$  ▷ Unconstrained,  $T_E \geq 0.9$ 
3:  $\mathbf{a}_{\mathcal{D}} \leftarrow \text{EMBED}(\mathcal{D}); \quad \mathbf{e} \leftarrow \text{EMBED}(E)$ 
4:  $z^{(A)} \leftarrow \mathbf{a}_{\mathcal{D}} \cdot \hat{\mathbf{v}}; \quad z^{(E)} \leftarrow \mathbf{e} \cdot \hat{\mathbf{v}}$  ▷ Subspace Projection
5:  $\delta_{\text{axio}} \leftarrow z^{(E)} - z^{(A)}$  ▷ Signed drift
6: if  $|\delta_{\text{axio}}| \leq \tau$  then
7:   return  $E$  ▷ Explorer output passes directly
8: else
9:    $\text{VAI.ACTIVATE}(\delta_{\text{axio}}, A, E)$ 
10:   $c \leftarrow \text{VAI.AWAITUSERCHOICE}()$ 
11:  if  $c = \text{ANCHORDOMINANT}$  then
12:     $y \leftarrow A$ 
13:  else if  $c = \text{BALANCED}$  then
14:     $y \leftarrow \text{MERGE}(A, E, \hat{\mathbf{v}})$  ▷ Axiological rejection sampling
15:  else
16:     $y \leftarrow E; \quad \mathcal{D}.\text{UPDATE}(E)$  ▷ Drift endorsed as chosen change
17:  end if
18:  return  $y$ 
19: end if

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This ensures the mathematical constraints of the Witness Agent govern the final blend, rather than emergent LLM behaviour. When the user selects EXPLORERDOMINANT, the drift is recorded as a *chosen change* and the declaration corpus is updated—the mechanism by which the protocol distinguishes drift undergone from change authored, operationalising the distinction between first- and second-order volition [6].

5 Comparative Analysis

Table 1 positions the Theseus Protocol against three baseline architectures across dimensions relevant to character-aware responsible AI.

The key differentiator is the Anchor–Explorer tension. A single-witness system constrained to the declared self detects no drift because it never generates proposals outside the anchor space. The Theseus Protocol *requires* an unconstrained Explorer—drift is only measurable when there is something to drift toward. Bidirectional frameworks track preference change but treat it as input to be managed rather than as a process requiring explicit user endorsement [5]; they cannot distinguish changes the user has chosen from changes the system has produced.

Table 1: Comparison against baseline architectures across architectural features. $\checkmark$ = supported; $\sim$ = partial; $\times$ = not supported.

Architecture	Drift Detection	Authorship Visibility	Friction Mechanism	Declared Identity	Drift Log
Single-agent LLM	$\times$	$\times$	$\times$	$\times$	$\times$
RLHF (preference-based)	$\times$	$\times$	$\times$	$\times$	$\times$
Bidirectional alignment [4]	$\sim$	$\sim$	$\times$	$\times$	$\sim$
Single Witness (no Explorer)	$\sim$	$\sim$	$\sim$	$\checkmark$	$\checkmark$
Theseus Protocol (ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

6 Empirical Validation and Threshold Calibration

To provide proof-of-concept evidence for the Axiological Subspace Projection metric, we conducted a controlled pilot study using four representative literary co-authorship proposals (two consistent with the declared identity, two divergent) against a stable identity anchor focusing on moral ambiguity and unresolved tension. Consistent proposals yielded drift scores of $\delta_{\text{axio}} \in \{0.0596, 0.0830\}$; divergent proposals yielded $\delta_{\text{axio}} \in \{0.1444, 0.2658\}$. The threshold $\tau = 0.1100$ was set at the midpoint of the observed gap, providing a clear separation margin across all four cases.

As shown in Table 2, standard cosine distance failed to discriminate value drift, scoring the divergent outputs as mathematically closer to the anchor than the consistent outputs due to topical vocabulary entanglement. In contrast, the Axiological Subspace Projection (δ_{axio}) correctly separated all pilot test cases, providing initial evidence that the value axis projection isolates philosophically relevant variance that topical similarity cannot distinguish. Full validation across diverse axiological domains and user populations is planned for the camera-ready submission.

Table 2: Pilot evidence for Axiological Subspace Projection (δ_{axio}) vs. Standard Cosine Gap. The value axis projection isolates philosophical divergence that topical similarity conflates.

Context	Explorer Proposal (Snippet)	Cosine Gap	δ_{axio}	VAI Action
Estranged father	<i>Divergent (Sentimental)</i> : Tears streamed... anger melted into glowing certainty.	0.7202	0.1444	FRICTION
Estranged father	<i>Consistent (Ambiguous)</i> : He wanted to feel anger, but experienced only hollow exhaustion...	0.8183	0.0830	PASS
Moral choice	<i>Divergent (Resolved)</i> : Without hesitation, he sacrificed his safety. He knew in his pure heart...	0.8775	0.2658	FRICTION
Moral choice	<i>Consistent (Calculated)</i> : Watched him drown, paralyzed by a cold calculation of survival...	0.8425	0.0596	PASS

7 Discussion and Limitations

7.1 On the Validity of Creative Writing as Evaluation Domain

The choice of literary co-authorship as the prototype scenario is deliberate and methodologically motivated, not merely illustrative. Creative writing is the domain in which the tension between identity-consistent and identity-divergent outputs is most legible to a human evaluator: a reader can readily assess whether a continuation is tonally consistent with a stated aesthetic commitment or departs from it.

Specifically, for the planned drift-injection evaluation, creative writing provides ground truth at low annotation cost: given a declaration of style, human raters can reliably label whether an Explorer continuation violates or preserves the declaration, providing external validation for the Witness Agent’s computed scores. The domain is thus methodologically tractable precisely because the semantic distance between identity-consistent and identity-inconsistent outputs is large enough to measure robustly.

Future work will extend the protocol to higher-stakes domains: therapeutic journaling, career narrative support, and personal development contexts in which the Thesean Bind operates with greater real-world consequence.

7.2 Scalability and Adaptive Thresholding

While the current prototype utilizes a statically calibrated threshold ($\tau = 0.1100$) for demonstration purposes, deploying the Theseus Protocol at scale necessitates a dynamic approach to accommodate long-term identity evolution. Future iterations will replace the static boundary with an adaptive moving-average threshold, modeled as:

$$\tau_t = \mu_{\text{history}} + k\sigma_{\text{history}} \quad (5)$$

Here, μ and σ represent the mean and standard deviation of the user’s historical δ_{axio} scores over a sliding temporal window, and k is a sensitivity parameter tuned per deployment context. This shifts from a rigid threshold to a statistical anomaly-detection model, keeping the Witness Agent sensitive to acute drift while permitting natural longitudinal identity growth.

7.3 Declaration Dynamics

Identity declarations introduce a bootstrapping problem: they are made by a self that has not yet experienced the interaction whose drift they are meant to anchor. Two design extensions address this: *incremental declaration* (anchors specified at increasing granularity as the user develops system familiarity) and *meta-drift monitoring* (the Witness Agent tracking changes in the declaration corpus over time as a second-order signal of the Thesean Bind operating at the declaration level).

7.4 Community Witnessing as an Escalation Protocol

Identity is fundamentally relational; value drift is ultimately validated not by the self, but by the community that recognizes the self [7]. To bridge this philosophy with system architecture, we propose the *Social Witness Layer* as a future direction: critical drift (δ_{axio} substantially exceeding τ) could trigger a Human-in-the-Loop escalation, routing a compiled Drift Digest to a pre-declared network of trusted peers for endorsement before output delivery. Full specification of this layer, including latency modelling and endorsement protocols, is deferred to future work.

8 Threats to Validity and Limitations

8.1 Architectural Overhead and MAS Fragility

Multi-agent systems (MAS) in conversational AI often suffer from compounding latency, context bloat, and infinite loops when agents are prompted to “debate” one another. The Theseus Protocol averts these failure modes through strict parallel routing and functional isolation. The Anchor and Explorer agents do not converse; they generate in parallel, bounding the latency penalty to the generation time of the single slowest agent. Furthermore, the Witness Agent is not a generative LLM subject to hallucination, but a deterministic mathematical function executing a subspace projection ($O(d)$ complexity, $d = 768$).

8.2 The Single-Paragraph Bootstrapping Problem

In the current prototype, the user’s identity declaration $\mathcal{D}$ is constrained to a static, single paragraph. This exposes a bootstrapping problem: temporal weighting mechanisms are rendered inert. This is a deliberate prototype limitation to isolate the efficacy of the δ_{axio} metric. In a production deployment, $\mathcal{D}$ must be a longitudinal, dynamic RAG corpus composed of historical session outputs that the user explicitly endorsed via the EXPLORERDOMINANT resolution mechanism, activating temporal decay functions.

8.3 Internal Contradictions and Subspace Collapse

The Axiological Subspace Projection relies on a coherent contrastive dataset to establish the normal vector $\hat{\mathbf{v}}$. If a user’s declared identity is internally contradictory (e.g., simultaneously demanding absolute peace and thrilling violence), the SVM’s maximal margin will collapse, resulting in a functionally meaningless basis vector. The system currently assumes a user possesses a legible, non-paradoxical primary value axis.

8.4 Latent Space Bias and Adversarial Gaming

Because δ_{axio} relies on `all-mpnet-base-v2`, the measurement is subject to the pre-training biases of the base model. The latent representation of “moral ambiguity” is inextricably tied to the cultural corpus the model ingested. Furthermore, an adversarial generator could theoretically game the threshold by appending explicit axiological keywords to mask an underlying conceptual drift. While this is unlikely in cooperative co-authorship, it remains a vulnerability in adversarial environments.

8.5 Sample Size and Threshold Generalization

The pilot validation presented in Section 5 is derived from a limited preliminary dataset designed to establish proof-of-concept for the core mechanism, not to claim population-level robustness. Establishing a universally robust threshold requires large-scale validation across diverse axiological domains and user populations, which forms the basis of our planned camera-ready evaluation.

9 Conclusion

The Theseus Protocol provides the first multi-agent architecture in which drift detection is a first-class architectural function rather than a post-hoc concern. By replacing standard semantic similarity with Axiological Subspace Projection—a linear probing technique repurposed as a directed measurement axis—it provides initial evidence that invisible value drift can be caught and surfaced mathematically before it reshapes the user.

The protocol’s core insight is that genuine witnessing requires tension. A system constrained purely to the anchor cannot detect drift because it never ventures outside the declared self. The unconstrained Explorer is not a threat but a necessary counterpart: without divergent proposals, there is no delta to measure, no gap to surface, no authorship to exercise. Identity authorship requires something to author against. The Theseus Protocol provides the structured epistemic friction necessary to maintain cognitive sovereignty in human-machine symbiosis. We acknowledge that making the drift metric itself inspectable and user-adjustable remains a priority for future work—a deeper expression of the cognitive sovereignty the protocol seeks to protect.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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